

Review 02 — NA convergence & apodization

mcdo_dev Phase 2 (output/02_na_apodization/
docs/theory/02_apodization_scalar_debye.md)

2026-06-12

Modules and theory chain

apodization.py: aplanatic Gaussian illumination (sine condition $r = f \sin \theta$)

$$A(\theta) = \exp\left(-\alpha_t \frac{\sin^2 \theta}{\sin^2 \alpha}\right), \quad \alpha_t = \left(\frac{r_{ap}}{w}\right)^2 \quad [\text{Horváth \& Bor 2003}]$$

debye.py: scalar Debye integral (same $A(\theta)$ and $\sqrt{\cos \theta}$, vector kernels dropped) + analytic Airy $[2J_1(v)/v]^2$ and axial $\text{sinc}^2(u/4)$.

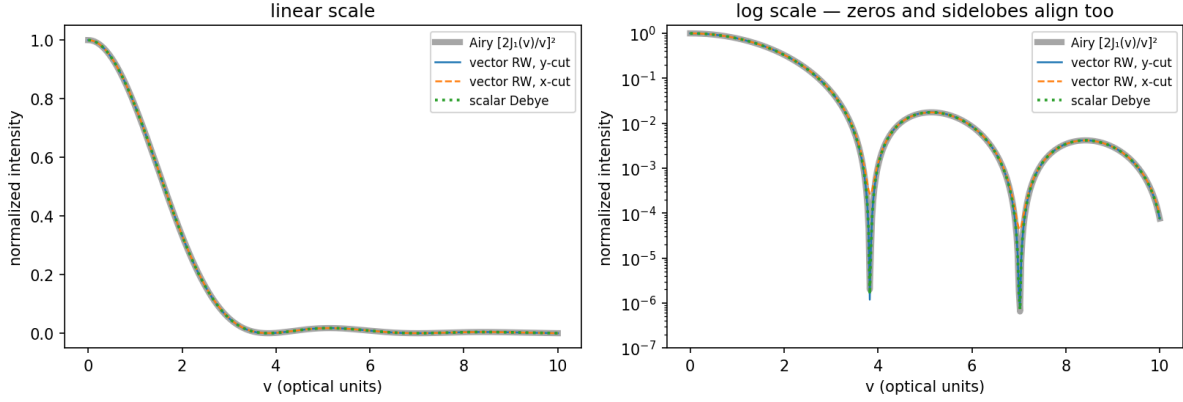
Chain logic: RW – scalar isolates *polarization (vector) effects*; scalar – Airy isolates *wide-angle geometry*. All results at $\lambda = 750 \text{ nm}$, $n = 1.3$, $X_{NA} = n \sin \alpha$.

Key numbers

- Low-NA correctness: all theories coincide with Airy to 10^{-4} – 10^{-3} at $X_{NA} = 0.1$ (check 1).
- Vector thresholds (uniform): x -cut vs scalar exceeds 1% at $X_{NA} \approx 0.30$, 5% at ≈ 0.60 ; y -cut stays $\lesssim 2\%$ to ≈ 0.9 (check 3) — threshold statements require polarization + cut.
- Axial first minimum: $u_0 4\pi \rightarrow 8.98$ and depth $10^{-7} \rightarrow 10^{-2}$ for $X_{NA} 0.1 \rightarrow 1.2$ (check 4).
- Apodization: $\alpha_t \rightarrow 0$ regression = uniform to 10^{-13} ; $\text{FWHM}_v 3.17 \rightarrow 4.80$ for $\alpha_t 0 \rightarrow 4$, Airy rings suppressed (check 5).
- Anisotropy = longitudinal field: $\text{FWHM}_x/\text{FWHM}_y \rightarrow 1.4$ at $X_{NA} = 1.2$, tracking the E_z energy fraction (check 6).
- Axial direction (check 7): scalar holds far longer on axis — uniform reaches 1% only at $X_{NA} \approx 1.15$ (never 5%); the on-axis vector ingredient is just the smooth $(1 + \cos \theta)$ kernel (no E_z , no anisotropy). With $\alpha_t=4$ the axial deviation grows earlier (1% @ 0.60) — kernel reweighting moves a soft pupil more than a hard one.
- Input polarization (check 8; headline results stay linear- x): circular input gives $|I_0|^2 + |I_2|^2 + 2|I_1|^2$ (no zeros); its thresholds are 1%/5% at $X_{NA} \approx 0.45/0.90$ — the old report’s lenient numbers recovered under its (circular) convention.
- **Corrected threshold study (reverses the old progress report):** with the scalar reference using the *same* $A(\theta)$, Gaussian illumination *extends* scalar validity (table below). The report’s “Gaussian fails at $\text{NA} \approx 0.3$ ” compared against the paraxial Gaussian-beam formula (invalid at high NA); its lenient uniform numbers came from circular-polarization averaging. Errata: docs/report_errata.md.

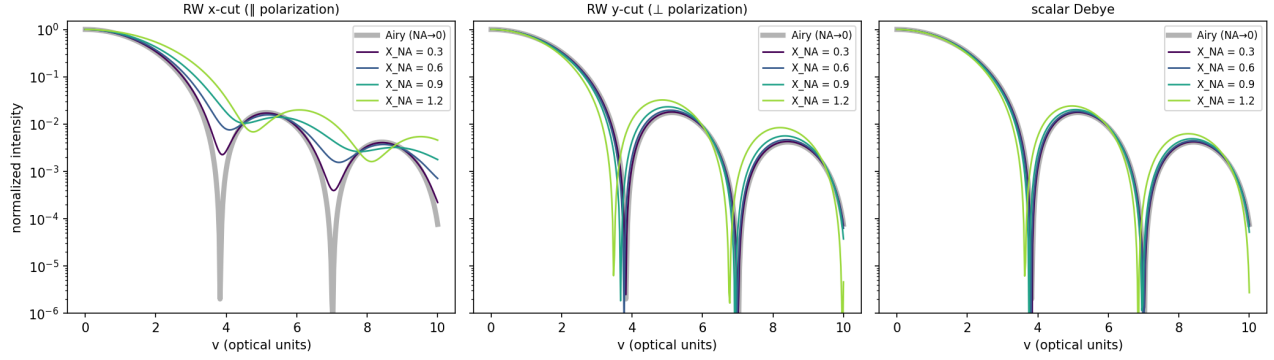
illumination	x -cut 1%	x -cut 5%	x -dev at $\sin \alpha = 0.9$
uniform	$X_{NA} 0.30$	0.60	32.1%
Gaussian $\alpha_t=1$	0.35	0.65	25.4%
Gaussian $\alpha_t=4$	0.40	0.85	12.6%

Check 1 — $X_{NA} = 0.1$: every theory must coincide with the Airy analytic



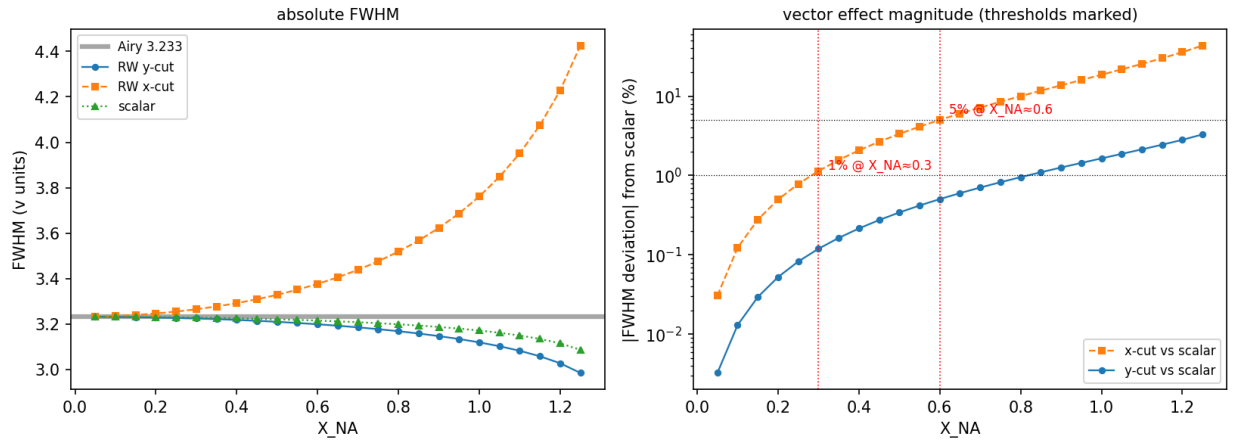
Check 1 (low-NA correctness). $X_{NA} = 0.1$: vector RW (both cuts), scalar Debye and analytic Airy coincide; the log panel shows zeros and sidelobes aligned down to 10^{-6} .

Check 2 — how each cut evolves with NA (x-cut broadens & loses zeros; y-cut narrows, keeps zeros; scalar barely moves)

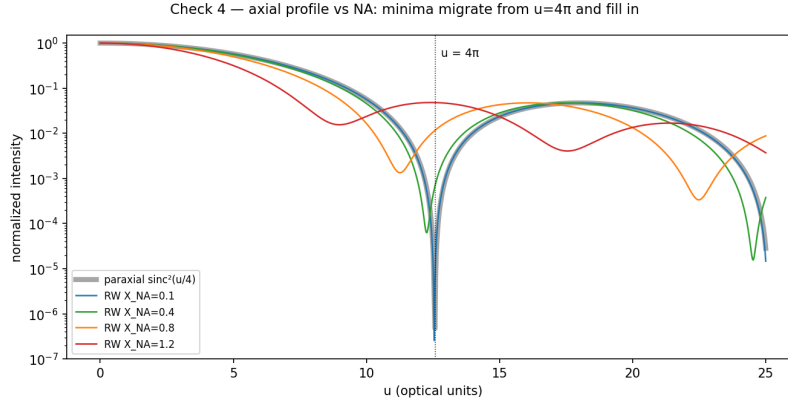


Check 2 (high-NA behavior). NA line-families per cut: the x -cut's zeros fill (longitudinal E_z) and its lobe broadens with NA; the y -cut narrows and keeps true zeros; scalar barely moves.

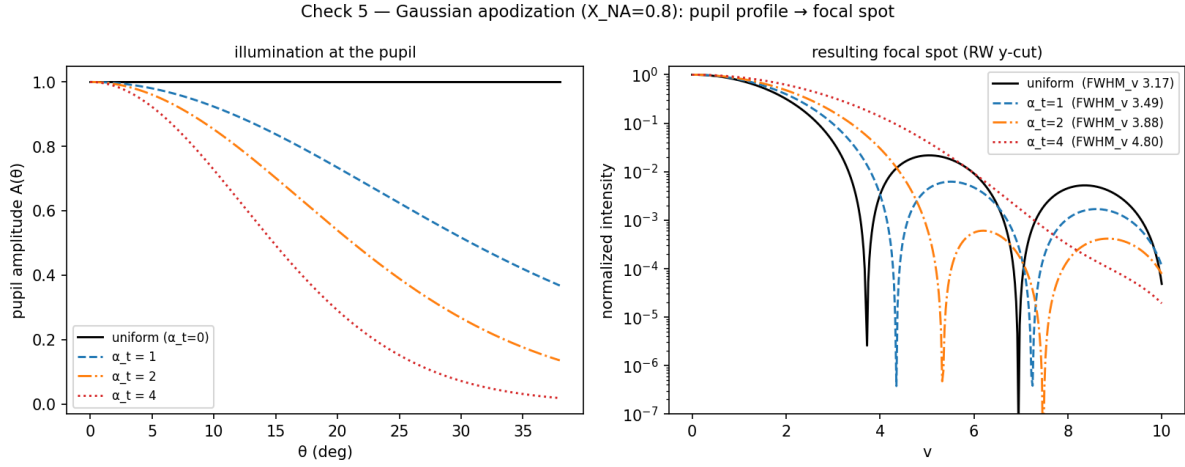
Check 3 — focal-spot FWHM vs NA: where is vector theory necessary?



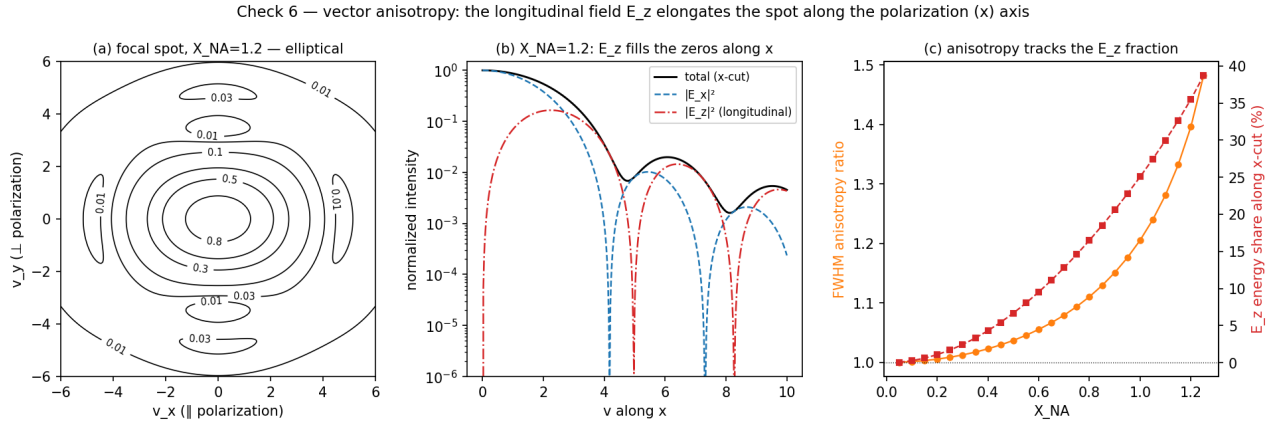
Check 3 (thresholds, uniform). FWHM vs NA and %-deviation with the 1%/5% crossings marked at $X_{NA} \approx 0.30/0.60$ (x -cut).



Check 4 (axial). Minima migrate focus-ward from $u = 4\pi$ and fill in at high NA — axial zeros are not exact (relevant to optical sectioning).

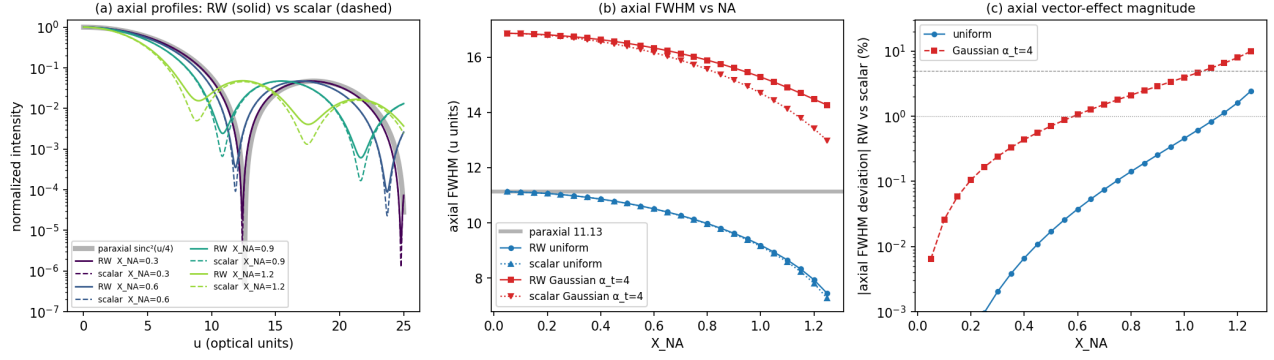


Check 5 (apodization). Pupil profile $\rightarrow$ focal spot: underfilling ($\alpha_t \uparrow$) broadens the spot and suppresses the hard-edge Airy rings.



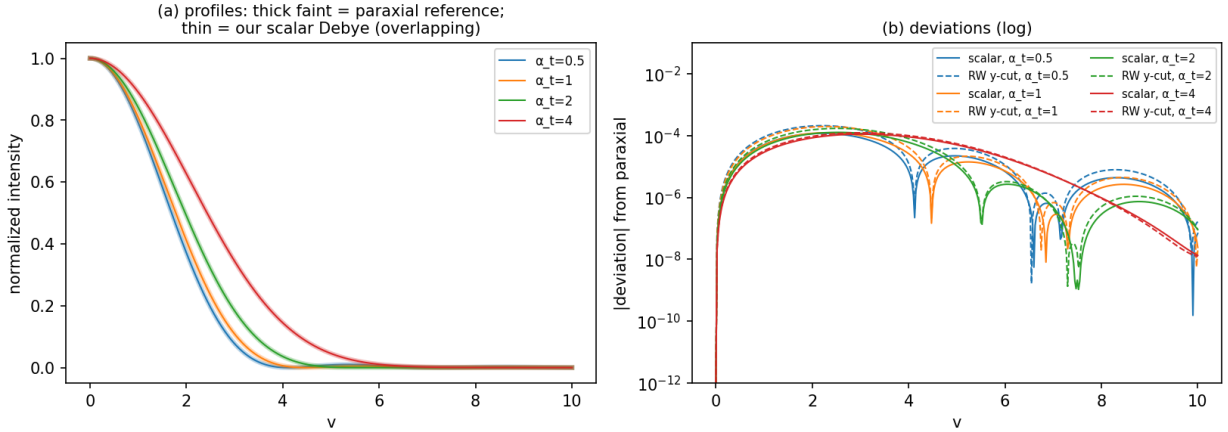
Check 6 (anisotropy). Elliptical spot at $X_{NA} = 1.2$; along x the longitudinal $|E_z|^2$ fills the $|E_x|^2$ zeros; the FWHM anisotropy ratio tracks the E_z energy fraction.

Check 7 — axial direction: RW vs scalar Debye vs NA (on-axis the vector effect is the $(1+\cos\theta)$ kernel; no x/y anisotropy)



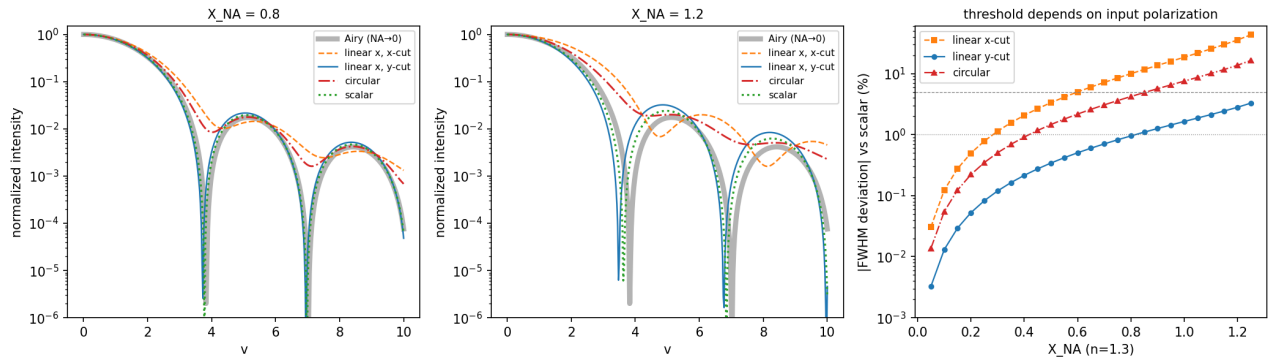
Check 7 (axial direction). RW vs scalar (same $A(\theta)$) along the axis: profiles per NA (a), axial FWHM vs NA (b), deviation (c). Uniform stays scalar-accurate to $X_{NA} \approx 1.15$ (1%); Gaussian $\alpha_t=4$ crosses 1% at 0.60. All curves are for a single linearly x -polarized input; on axis the polarization orientation is irrelevant.

Phase 2 closer — truncated-Gaussian focal profiles vs the paraxial reference (Tanaka/H&B regime, $X_{NA}=0.1$)



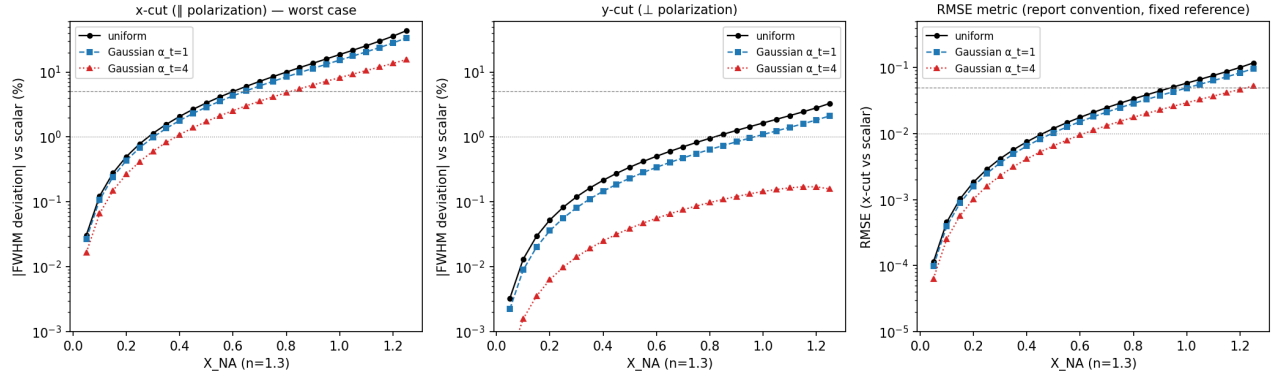
Closer — independent paraxial checkpoint. Truncated-Gaussian focal profiles vs a direct quadrature of $\int_0^1 e^{-\alpha_t \rho^2} J_0(v\rho) \rho d\rho$ (Tanaka 1985 / H&B regime, no shared code). Our scalar Debye *and* the RW y-cut match it to $\sim 10^{-4}$ for $\alpha_t=0.5-4$ at $X_{NA}=0.1$ — the apodization is correct against an external reference.

Check 8 — input polarization: linear- x cuts vs circular (circular = azimuthal combination; no zeros)



Check 8 (input polarization). Linear- x cuts vs circular input vs scalar at $X_{NA} = 0.8$ and 1.2 , and FWHM deviation vs NA per polarization: thresholds are 0.30/0.60 (linear, x -cut), 0.45/0.90 (circular), 0.85/- (linear, y -cut). Circular has no zeros and averages away the anisotropy.

Corrected NA-threshold study: RW vs scalar Debye with the SAME $A(\theta)$ (report slides 24–28 redo)



Corrected NA-threshold study (slides 24–28 redo): Gaussian $\alpha_t=4$ sits below uniform in every panel — x -cut, y -cut, and the report’s own RMSE metric. Gaussian illumination is *more* scalar-friendly, not less.